

CrossMEP: A Tiered Synthetic Dataset of Multi-Trade MEP Cross-Sections

A support-design problem distribution for learning-based structural support assembly synthesis

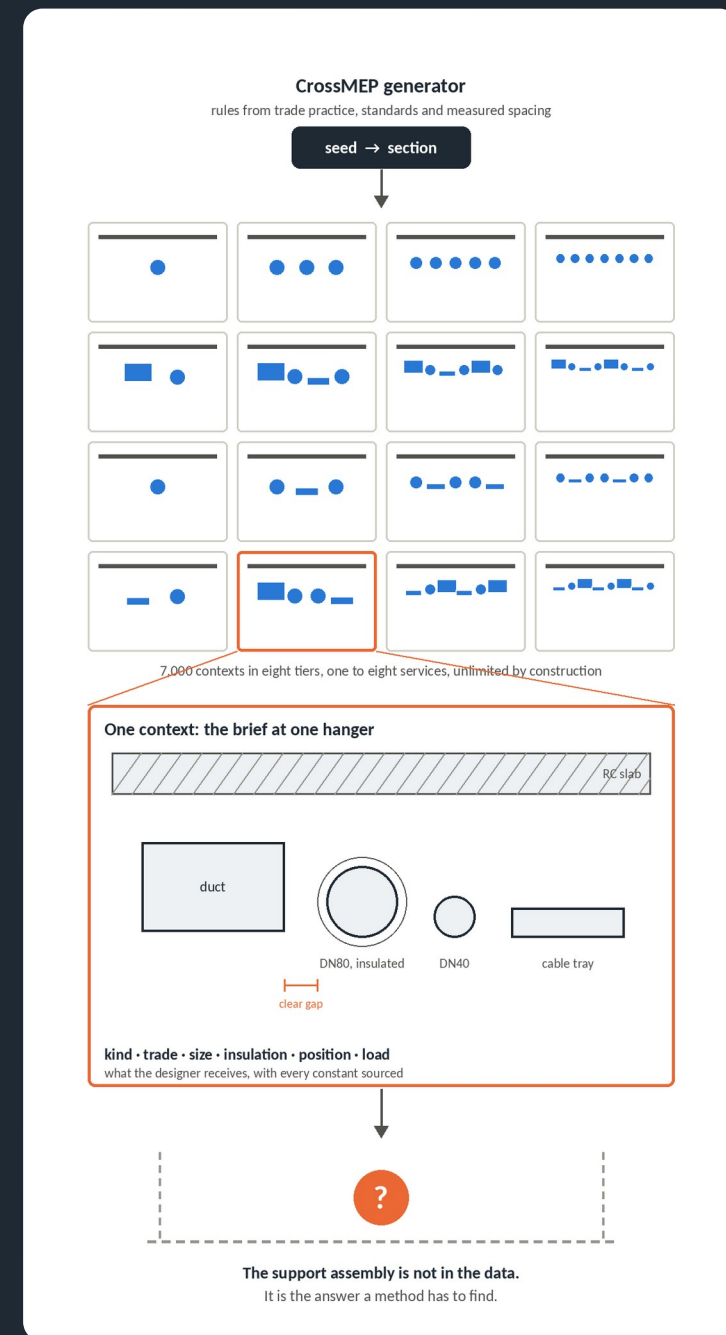
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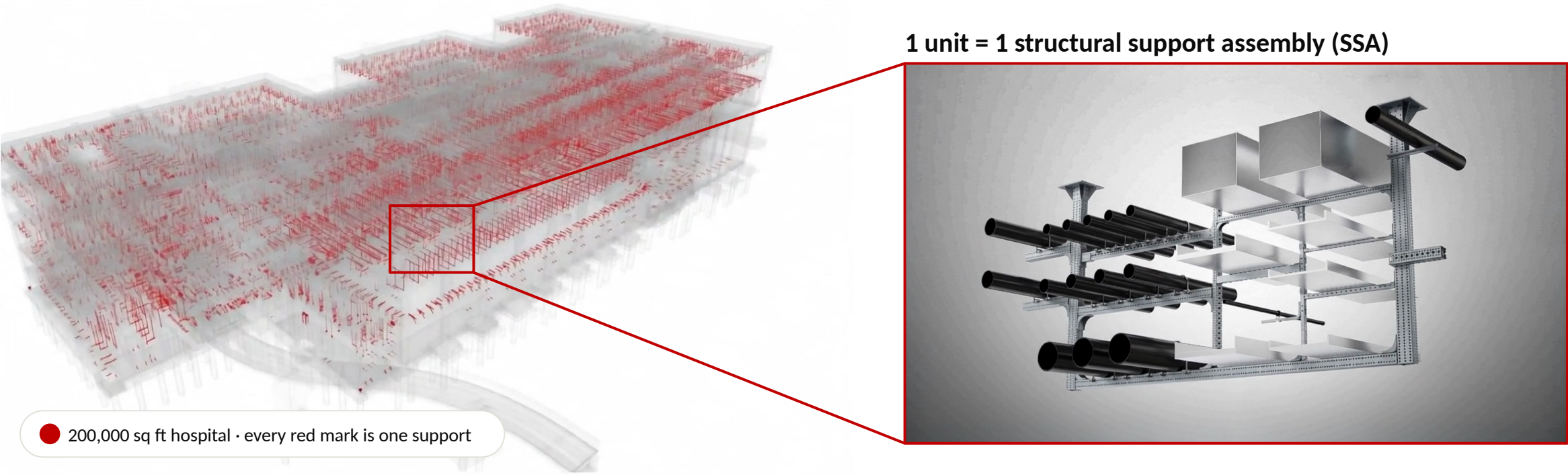
43rd International CIB W78 Conference • IT in Construction • New Delhi, 6–8 October 2026

Stanford

School of Engineering &
Doerr School of Sustainability
Civil & Environmental Engineering



The synthesis of structural support assemblies (SSAs) takes 25% of MEP design effort because every assembly must be designed individually





≈ 10,000
support assemblies in one hospital project, each one its own design





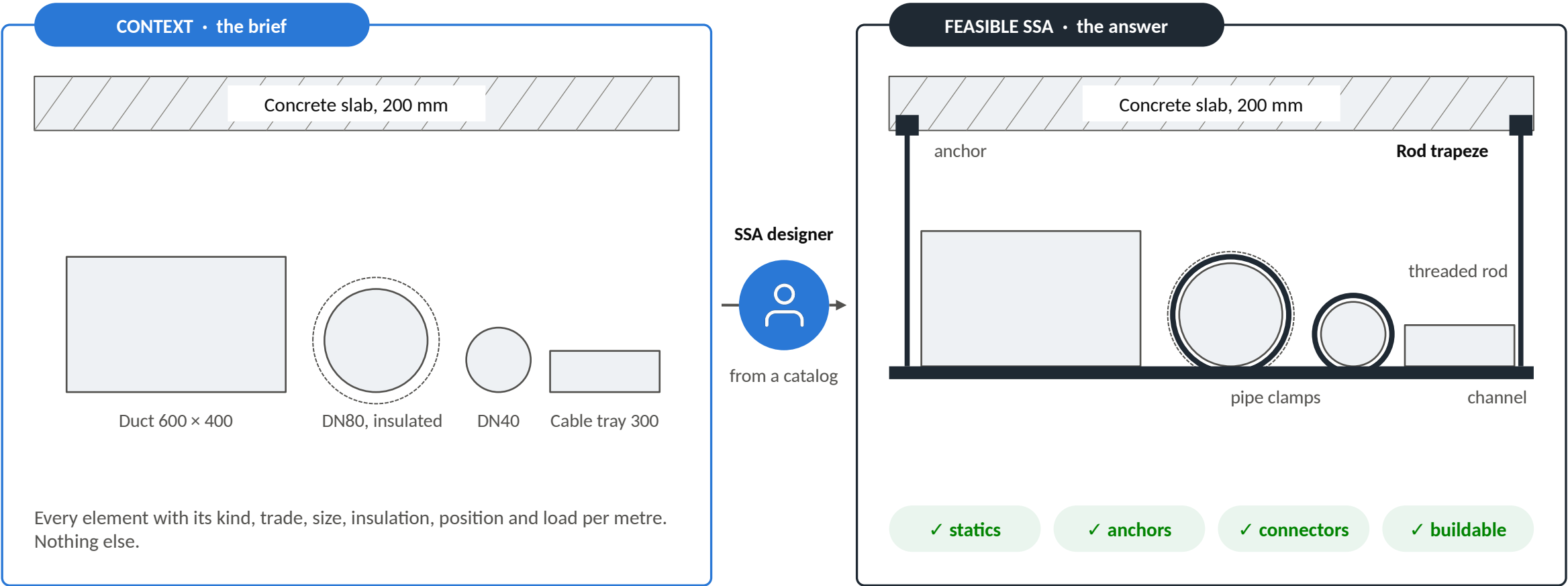
20 min – 2 h
of engineering time for each assembly, designed one by one



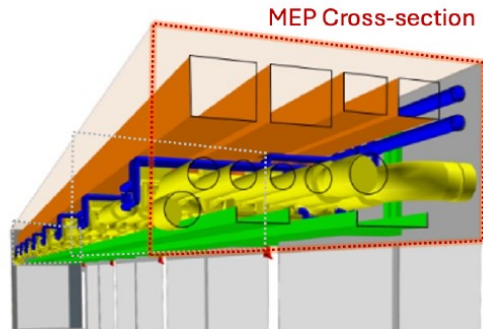


≈ 25 %
of MEP design effort, about \$600k of engineering cost per project

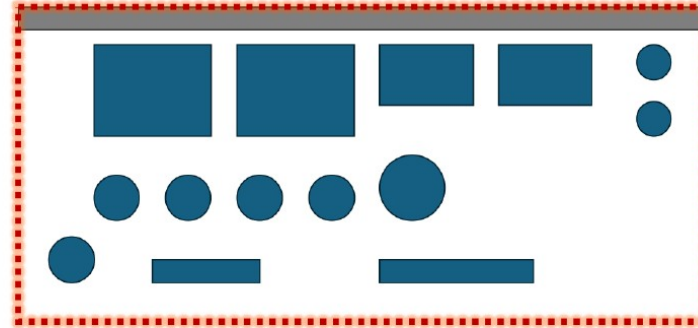
An SSA designer turns one MEP cross-section into a feasible structural support assembly



Existing tools coordinate and verify modular support assemblies, but their synthesis remains manual because it relies on tacit engineering expertise and complex catalog-driven rules



(a) Full MEP route



(b) One CrossMEP context

Design starts after coordination [3]: at each hanger the designer works from the section across the run, not from the whole model [2].



Where practice stands

Tools coordinate the model and check a design.

Choosing the topology and the parts, from a fixed catalog, is still done by hand, one section at a time [1].



1. Geometric judgment

Engineers choose an assembly's topology from experience.



2. Rule explosion

Code, load and material constraints interact across thousands of combinations.



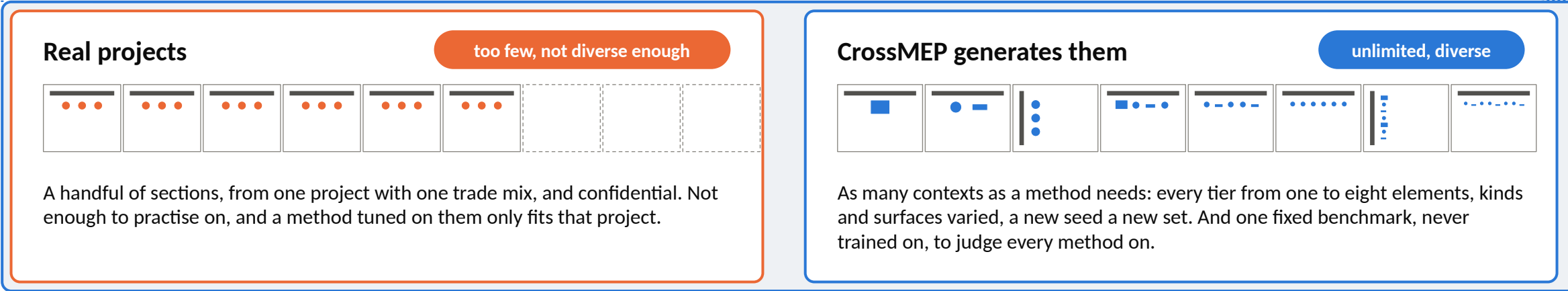
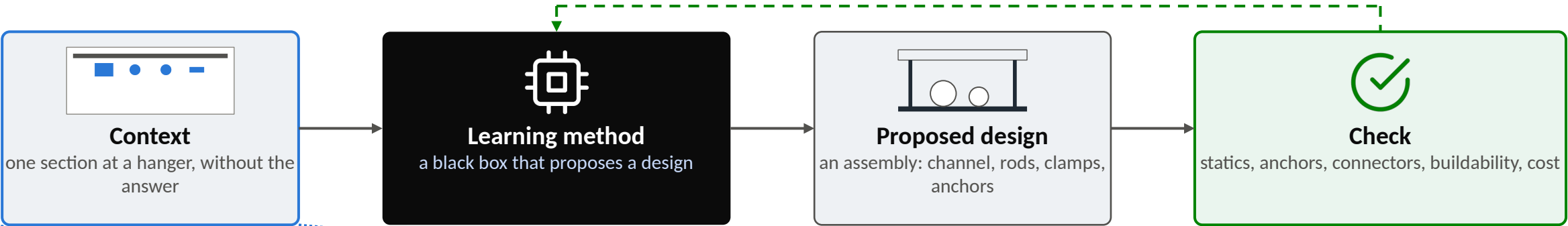
3. Catalog volatility

Catalogs change faster than rule-based systems can be rewritten.

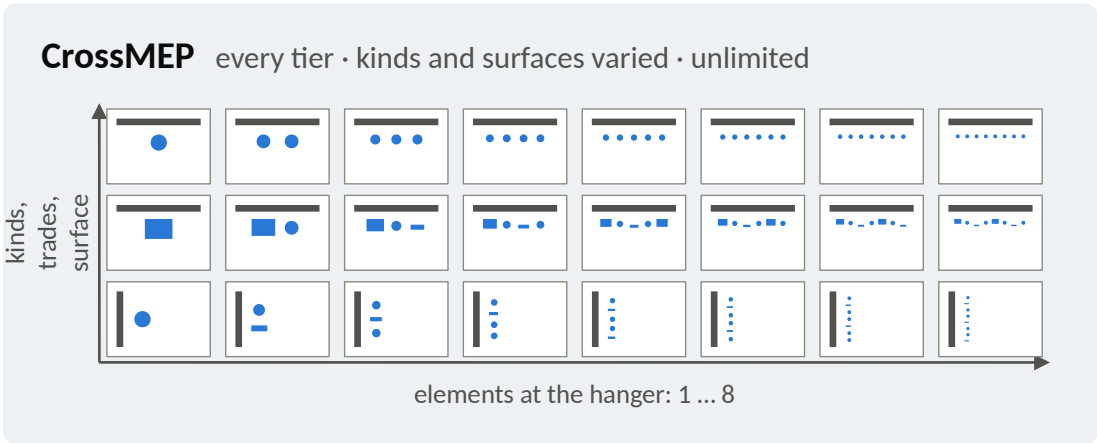
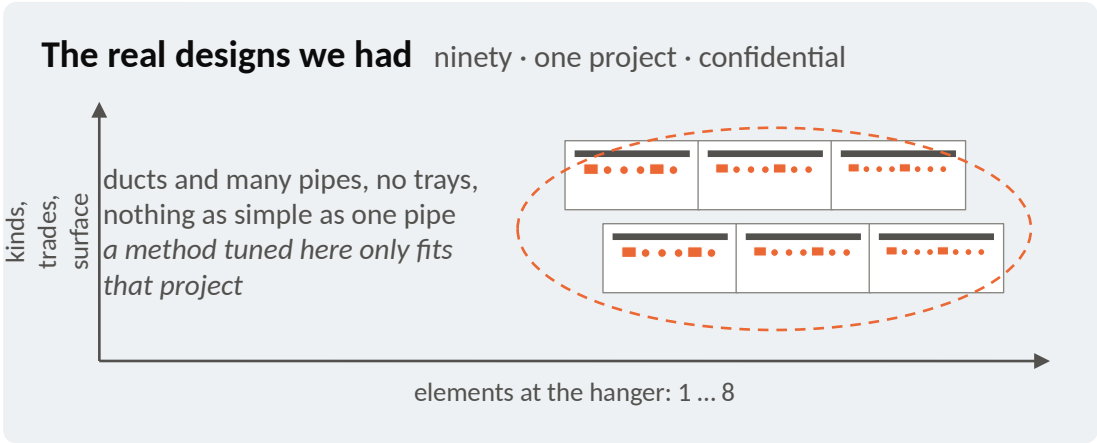
[1] Pedrollo, Graeber & Fischer (2026), Feasibility-aware sequential synthesis of structural support assemblies, ISARC 2026 [2] Pedrollo, Gvadzabia, Graeber & Fischer (2026), CrossMEP, CIB W78 2026 (this paper) [3] Korman, Fischer & Tatum (2003), Knowledge and reasoning for MEP coordination, J. Constr. Eng. Manage. 129(6)

A learning method needs thousands of contexts to practise on, and real projects give too few, so we generate them

How a learning method learns, round after round



The ninety real designs we had came from one project and filled one corner of the design space, so CrossMEP covers all of it by construction



CrossMEP at a glance



7,000

contexts in four splits, on disjoint seeds



31,484

elements: pipes, trays, ducts, conduits; six trades



C1–C8

eight tiers by exact element count



0 labels

the brief, not the answer



Traced

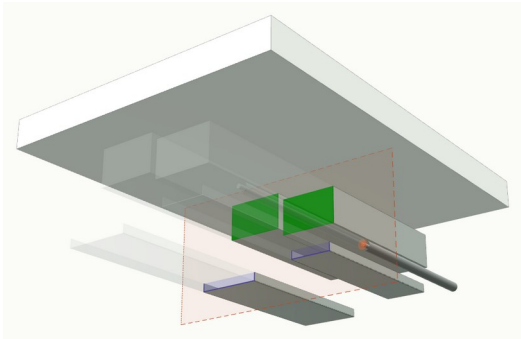
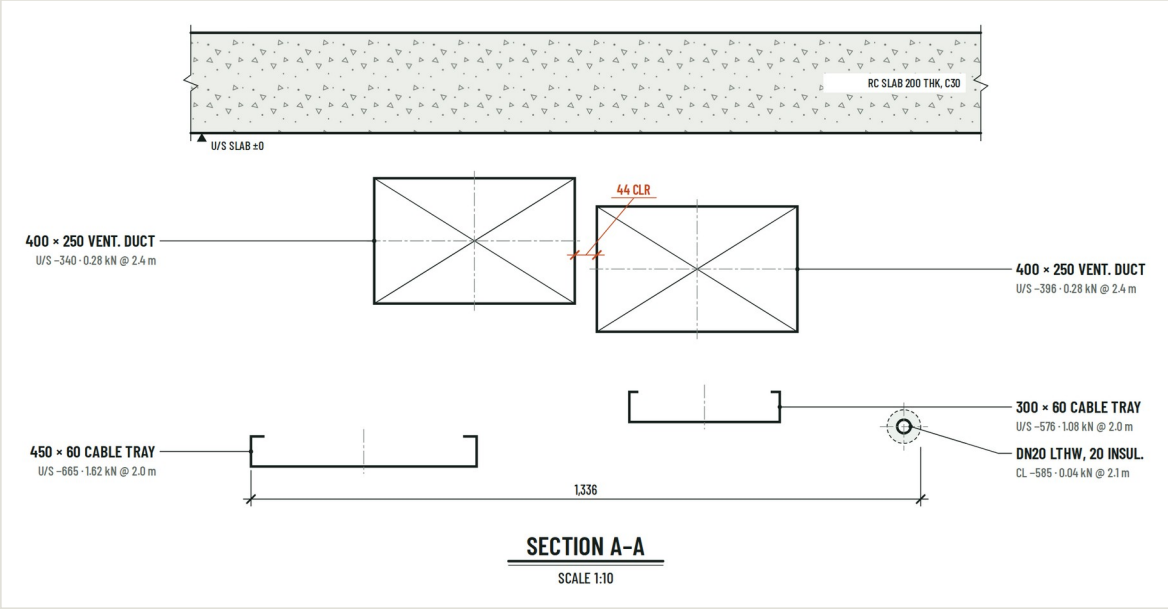
every constant sourced or declared



Open

data CC BY 4.0, code MIT, seeded generator

One context is exactly what a support designer receives at one hanger, without the answer



What every element carries

- kind
- service, trade
- size
- insulation
- position
- load = kN/m × span

● ventilation ● electrical ● heating

What the support designer is given

Element	Trade	Ins. mm	kN/m	Span	kN at support
400×250 duct	● ventilation	0	0.115	2.4 m	0.28
400×250 duct	● ventilation	0	0.115	2.4 m	0.28
450 tray	● electrical	0	0.809	2.0 m	1.62
300 tray	● electrical	0	0.540	2.0 m	1.08
DN20	● heating	20	0.019	2.1 m	0.04

**200 mm**
ceiling slab,
concrete C30

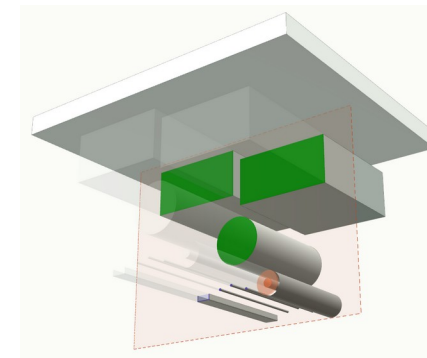
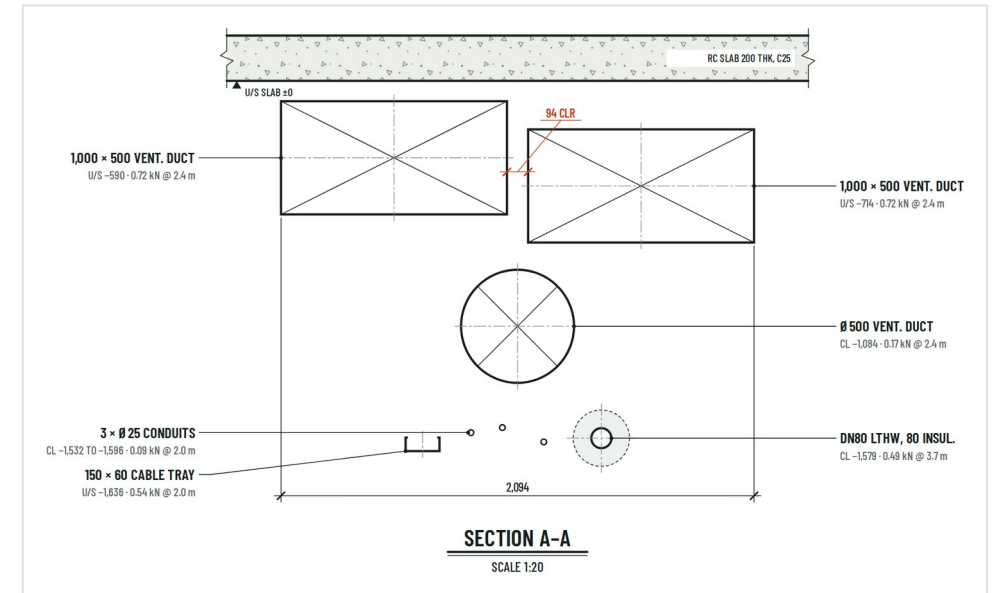
**3.30 kN**
total load at this
support

**44 mm**
closest clear gap,
2 rows

**Not given, by design:** channel, rods, clamps, anchors. No correct answer either: a feasible support depends on the catalog you build from.

Every section is built the way trades run services, with gaps measured on a real building

- 1 **# Tier**
Cn fixes the exact element count, $n = 1 \dots 8$
- 2 **Surface**
ceiling or wall (78 / 22 %); concrete, 150–300 mm thick
- 3 **Services**
grouped as trades run [3]: hot + cold, flow + return, conduit groups, 1–2 trays, ducts
- 4 **Rows**
bulky first: ducts nearest the slab, then trays and conduits, then pipes
- 5 **Spacing**
gaps drawn from a built project [12], at least 25 mm; electrical above wet on walls [10]
- 6 **Check**
every pair 25 mm clear, rows centred, everything recorded; same seed, same file



● ventilation ● electrical ● heating

One generated context, tier C8: 8 services on 3 rows, ducts nearest the slab, then the tray and conduits, then the pipe. The gap marked on the sheet is one draw from the measured distribution; the same seed gives the same section.

[3] Korman, Fischer & Tatum (2003), Knowledge and reasoning for MEP coordination, J. Constr. Eng. Manage. 129(6) [10] IET On-Site Guide (BS 7671), conduit support spacing; BS 7671 Reg. 528.3.2 [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

Every number is traced to a standard, to a measurement, or to a choice we declare as ours



Pipes DN15–150

Sizes EN 10220 / EN 10255 medium series [4]

Load steel + water × ASME B31.1 water-service span [8]



Insulation

Sizes GEG Anlage 8, heated lines [11]; 30 / 50 mm condensation control, chilled

Load adds to the size, not the load



Cable trays 150–600

Sizes IEC 61537 systems [6] in NEMA VE 1 widths [7]

Load 50 kg/m full at 300 mm × 2.0 m, below the shortest NEMA class span [7]



Ducts

Sizes EN 1505 rectangular, EN 1506 round [5]

Load manufacturer weight table with flanges × 2.4 m (SMACNA) [9]



Conduits Ø20–50

Sizes IEC 61386-1 [6], parallel groups of 2–6

Load steel tube + 40 % cable fill × 2.0 m (IET / BS 7671) [10]



Standards

what each element is and weighs: sizes, walls, spans, insulation [4]–[11]



Measured

how close neighbours sit: clear gaps and row stagger from two open buildings [12]



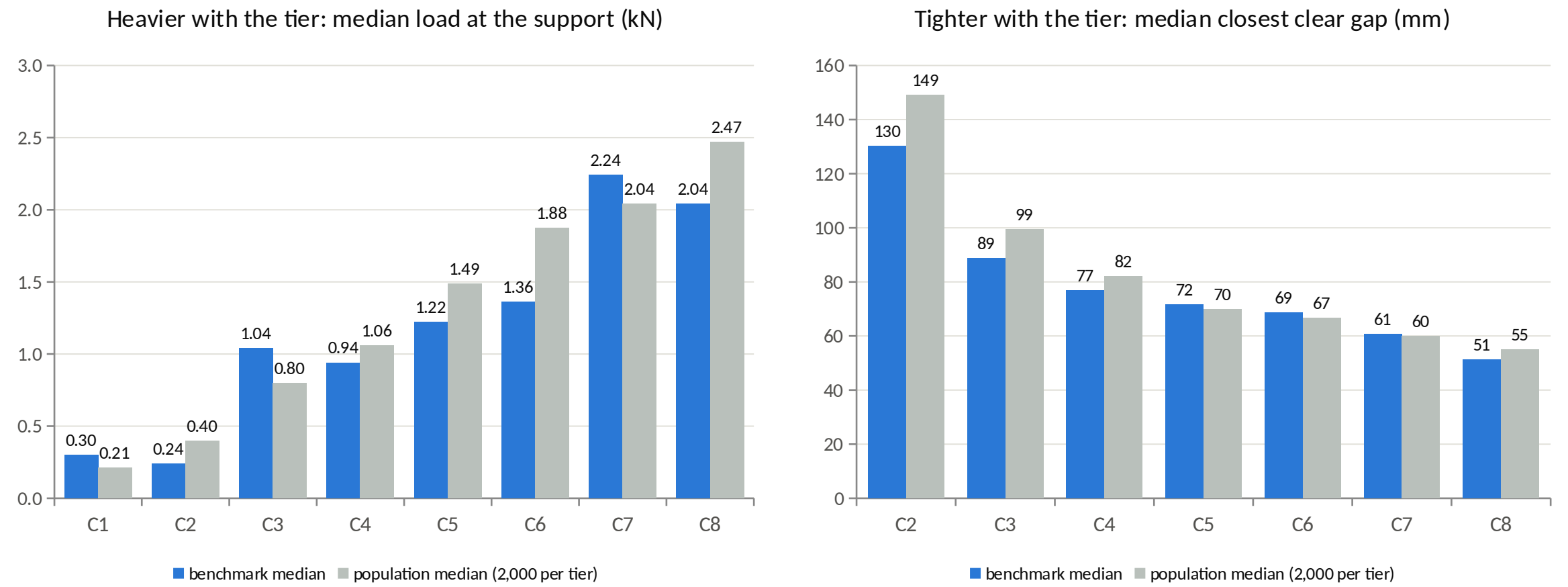
Declared

our choices, labelled as such: trade mix, ceiling / wall 78 / 22 %, number of rows

Every constant carries its source and its status, and why each standard was chosen over its alternatives is written down with it.

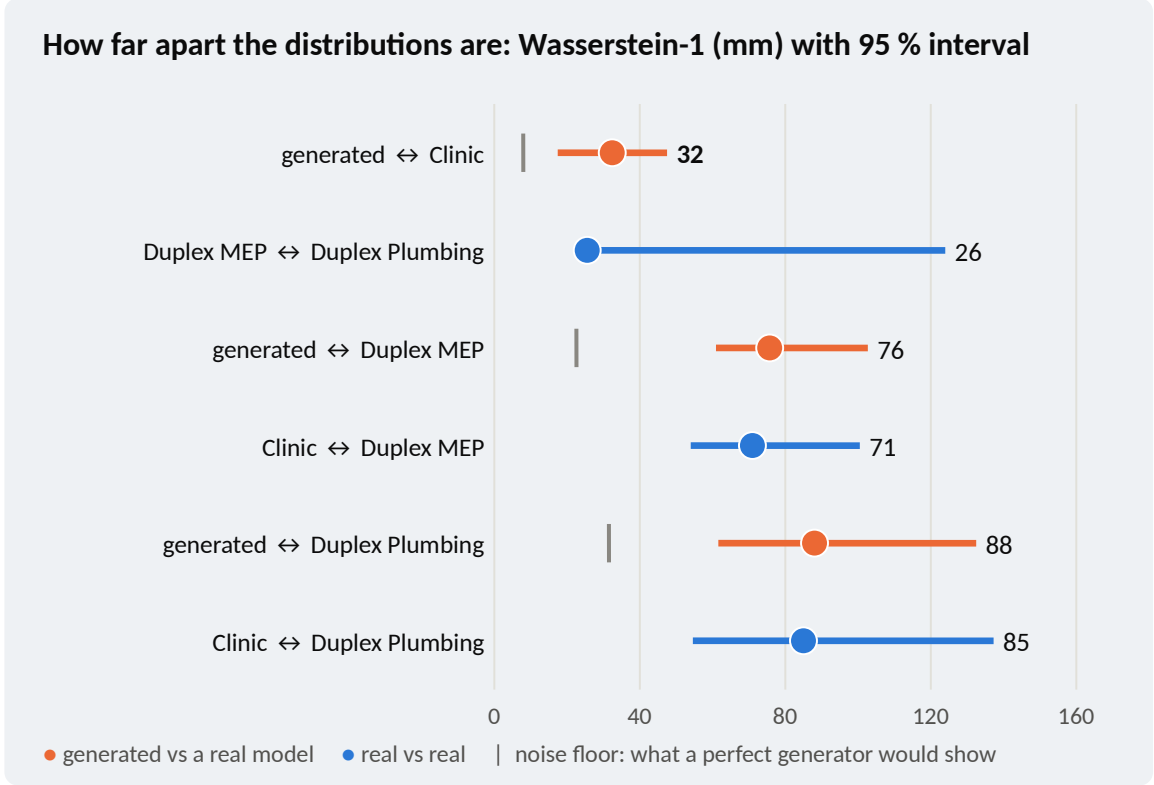
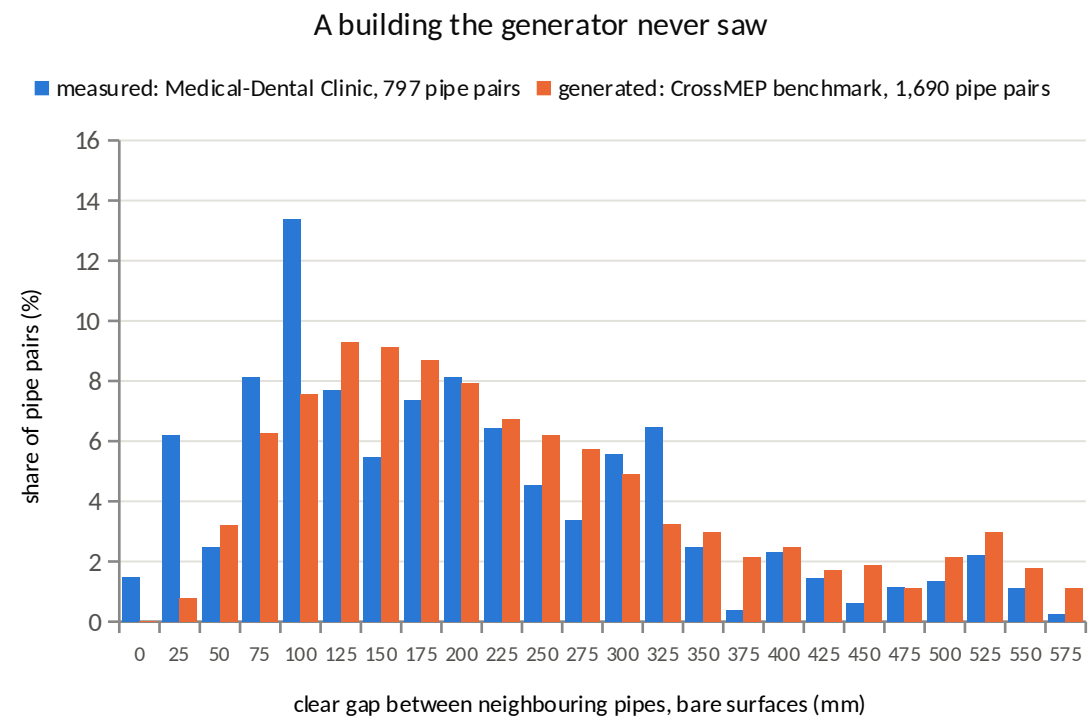
[4] EN 10220:2002, EN 10255:2004 (steel tubes: dimensions, medium series) [5] EN 1505:1997, EN 1506:2007 (sheet-metal air ducts: rectangular and circular dimensions) [6] IEC 61386-1 (conduit systems), IEC 61537 (cable tray and ladder systems) [7] NEMA VE 1 (metal cable tray systems: widths, load-class spans) [8] ASME B31.1 Table 121.5 (hanger spacing), as republished in the ASHRAE Handbook, HVAC Systems and Equipment [9] SMACNA, HVAC Duct Construction Standards, Metal and Flexible, Table 5-1 [10] IET On-Site Guide (BS 7671), conduit support spacing; BS 7671 Reg. 528.3.2 [11] GEG (Gebäudeenergiegesetz), Anlage 8 (pipe insulation thickness) [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

The tier fixes the element count, and load and congestion grow with it as the rules intend



A check that the dataset behaves as designed, not a discovery: the tier fixes the element count; load and congestion follow from the rules. Within a tier, kinds, trades, surfaces and stacking all vary. Load rises with the tier (rank correlation +0.46 on the benchmark, 125 contexts per tier; the population medians rise at every step) and the closest gap narrows at every step (−0.38); intervals are bootstrapped over contexts.

Generated spacing matches a clinic the generator never saw to within 32 millimetres



32 mm

generator to clinic (95 % CI 17-48); fitted on the duplex, clinic held out

26 mm

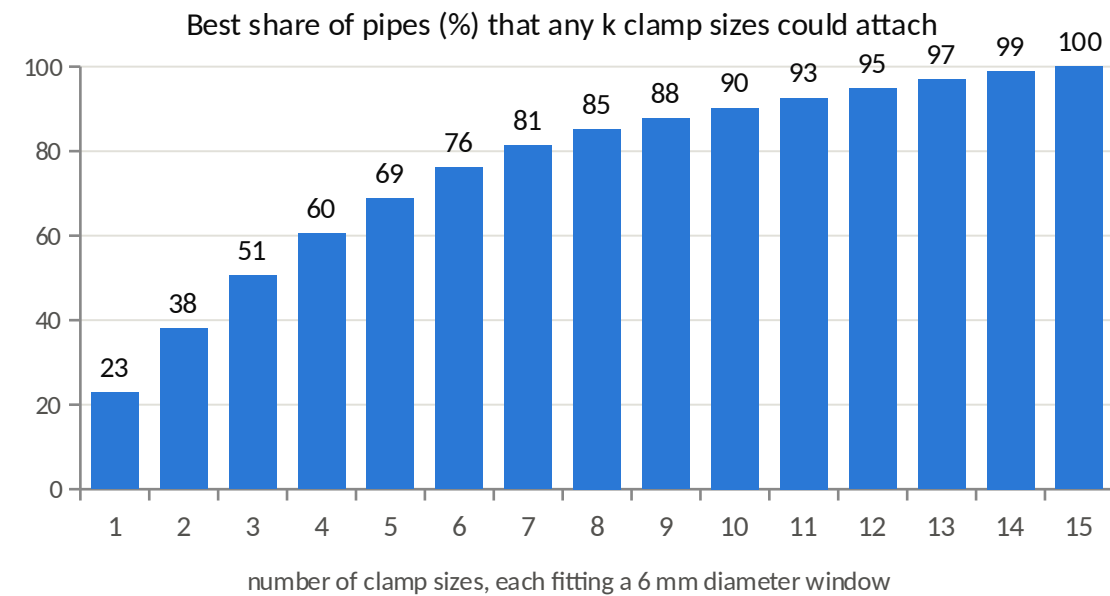
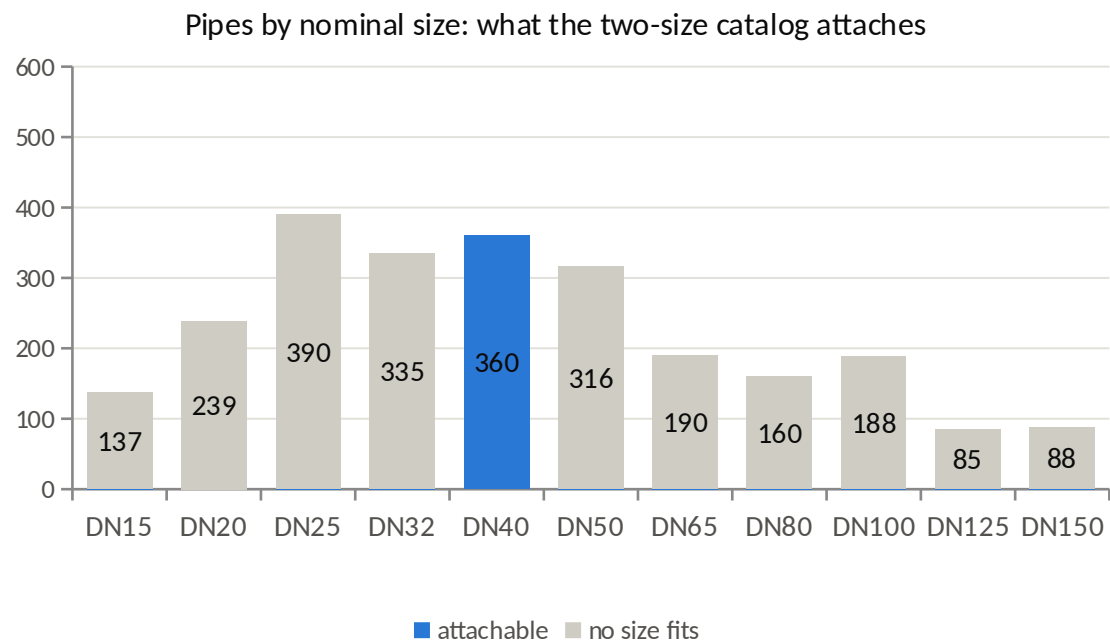
the duplex's own two models from each other: about as close as real is to real

190 mm

what a fixed 25 mm gap would be from the clinic

Sections every 250 mm, clinic pipe pairs weighted by shared length; models: buildingSMART Duplex Apartment and Medical-Dental Clinic [12]. [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

The paper's two clamp sizes reach one pipe in seven, and the dataset says which sizes to add



14.5 %

of benchmark pipes attachable with the paper's two sizes [2] (95 % CI 12.7–16.2); every one is DN40

38.1 %

with two sizes placed where the pipes are; 76 % with six, and every pipe with 15

0 of 2,012

trays, ducts and conduits: a pipe-clamp catalog defines no attachment. Load never binds (1.97 vs 2.5 kN)

[2] Pedrollo, Gvazdabia, Graeber & Fischer (2026), CrossMEP, CIB W78 2026 (this paper)

Pick a tier or an exact mix and a seed, and the Generator Studio returns the designer's brief

In: the parameters

SERVICES AT THE HANGER

By tier

Exact mix

TIER = NUMBER OF SERVICES

C1

C2

C3

C4

C5

C6

C7

C8

SEED

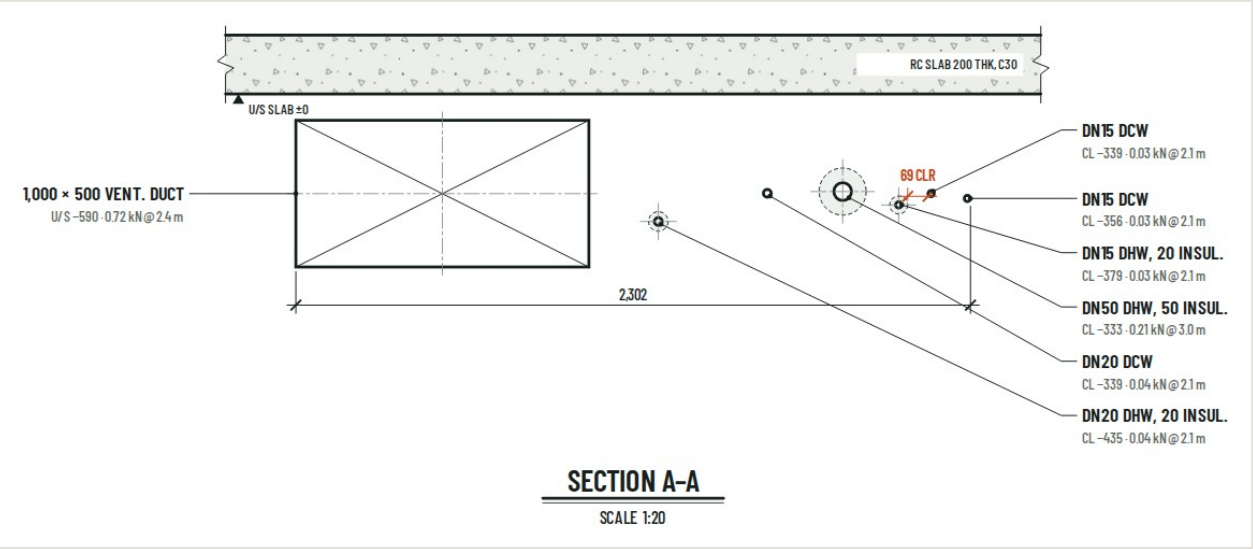
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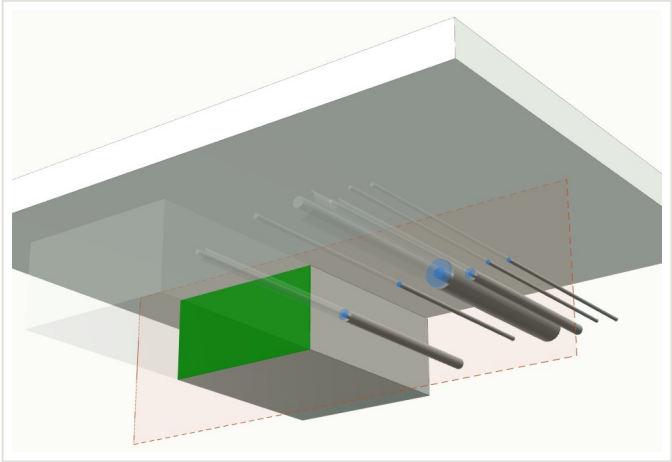
Generate another

Out: section A-A as the designer receives it, and what it carries



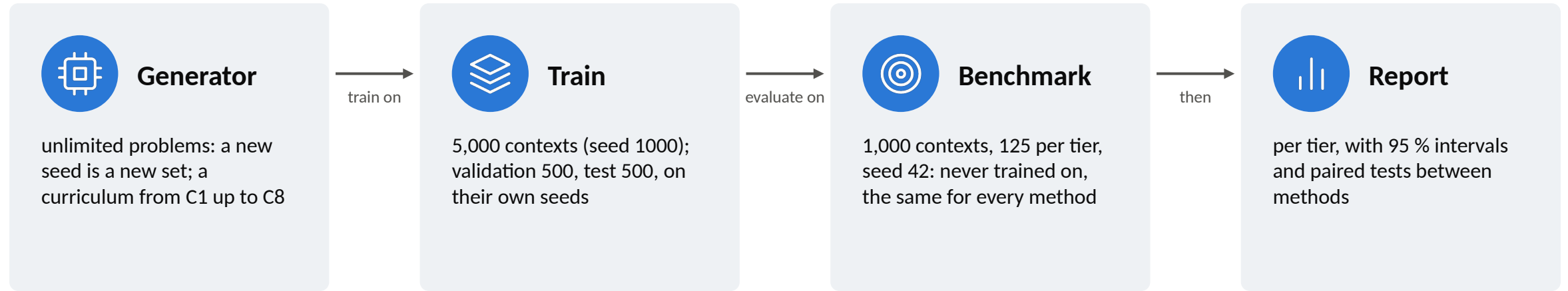
7 services in 1 row 1.10 kN at the support closest clear gap 69 mm hung from a 200 mm slab

Out: the same run in 3D, cut at the section



Scan to open the studio
every section is a stored output of the released generator, with the call that reproduces it

Methods train on the generator and are compared on one fixed benchmark, tier by tier



```
import crossmep.tasks as cm
data = cm.load("benchmark") # 1,000 contexts, or generate your own mix
from crossmep.evaluate import score, compare
score(my_results, data) # per tier, 95 % CI  compare(mine, baseline, data) # paired test
```

Four lines to load, score and compare; loading, metrics and scoring need only the Python standard library, the generator needs NumPy. The same files serve reinforcement learning, constraint programming and benchmarking: there are no labels to fit a method to.

Today CrossMEP is one section at one support, and next come a checker and real projects

Scope



One cross-section at one support: routing, branches and support spacing are outside it; the span is recorded, so a method can vary it



Realism checked on two open buildings, a residential duplex and a medical clinic; congested racks rest on practice and standards



Trade mix and tier composition are design choices, not survey results



Synthetic: not for the structural design of real installations

Next

1

A public checker for support designs, with best-known costs from an exact solver on the small tiers

2

The catalog as an input: generated catalogs, and catalogs a method has never seen

3

A real test set: sections from commercial projects with supports designed by engineers

If you coordinate services or design supports, open the studio, generate the sections you know from your projects, and tell us what looks wrong. Every rule is one documented constant away from being changed.



Support design finally has open problems to learn from

CrossMEP is the brief, not the answer: seven thousand support-design problems, open to anyone who wants to teach a machine to design supports.

Data CC BY 4.0, code MIT. Not for the structural design of real installations.

 **7,000**

support-design problems
in eight tiers

 **Sourced**

every constant sourced or
declared

 **Checked**

pipe gaps tested on two
open buildings

 **Open**

data, code and generator



github.com/lavinia-ped/CrossMEP_Dataset
lavinia@stanford.edu

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